Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-2026)
Faculty Name	Ms. G. Mamatha
Student Name	Brundavanam VR VR SrisaiHarini
Roll Number	2520030384
Branch	CSE
Course Outcome	CO-2

Case Study Topic: LibraryNet – Intelligent Digital Library & Knowledge Retrieval System

Work Done Summary:

In CO2, advanced indexing structures such as B-Trees and B+ Trees were implemented to manage large-scale digital library databases. Segment Trees and Fenwick Trees were used for analytics and cumulative borrowing statistics

Implementation Details:

B-Trees

- Used for indexing large book databases.
- Reduces disk access operations.
- Efficient for large datasets.

B+ Trees

- Supports range queries.
- Used for publication year and category searches.

Segment Trees

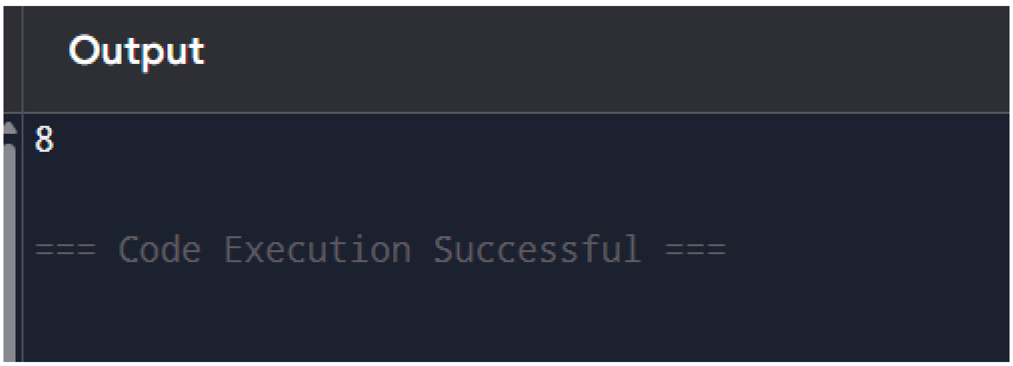
- Tracks borrowing trends across time intervals.
- Helps perform analytics efficiently.

Fenwick Trees

- Computes cumulative borrowing statistics.

Efficient for prefix sum calculations

Result : Screenshots / Output



Time Complexity

Data Structure	Operation	Time Complexity
B-Tree	Search	O(log n)
B+ Tree	Range Query	O(log n)
Segment Tree	Query	O(log n)
Fenwick Tree	Update	O(log n)
Fenwick Tree	Prefix Sum	O(log n)

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GitHub Link: https://github.com/HariniBrundavanam/DSA-2-project

Student Signature	**Faculty Signature**